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Theory and Practice of LLMs
Lab 3 Report

This report is a description of our solution to lab 3.
We will provide a bouquet of links that may be useful in understanding our experimentation, along with the chosen datasets and parameters for experimenting.
Each experiment or model will be followed by a commentary. The commentary is here to give a brief view of how we were thinking at each significant step of the experiment, the difficulties we have faced and our thought process in understanding and coming up with solutions.
The code will be provided on a github repository along with the necessary files.
Claude was used in creating the plots, and ChatGPT in cleaning up the blocks of code and for general presentation of the code.
For dataset cleanups we have reused code from other courses such as the Data Analysis and Big Data courses at Paris Cité. They have all been added to the Github repository.

For the scope of this study, we are looking at building a language model that is trained on French Poetry, and that will produce a poem of its own.

***In Experiment I, the training set is rather small, taken from Hugging Face*. (2.07 mb)**
In Experiment II, the training set is much larger (60 mb) taken from Malherbe Corpus. The cleaned datasets will be added to the repository.

IMPORTANT /// For the Github repository : Link

Now for our presentation…

(running locally
For Model 1, 2 and Experiment I: Intel core i9
For model 3 and Experiment II: cuda and GPU: Quadro RTX 3000
This has contributed significantly in the disparity of runtime performances)

Experiment I :

Hyperparameters *alpha* :

```
batch_size = 32
block_size = 256
max_iters = 5000
eval_interval = 500
eval_iters = 20
learning_rate = 3e-4
weight_decay = 0.01
```

Model 1 :

```
n_embd = 256
n_head = 2
n_layer = 3
dropout = 0.1
```

Output :
step 4999 | train loss 1.2513 | val loss 1.2740 | train bpc 1.8052 | val bpc 1.8380

*(Plotting was **not** done for **Model 1 and 2 (pre 5000 epochs)** because we **accidentally** deleted the powershell window containing the eval updates… but we know in a general sense that the starting loss is more or less the same across all Models. It would've been interesting however for us to plot the variation between the training and validation loss on these early experimentations. **We will do it** for later Models however).*

--- SAMPLE ---

<POEM=La Vérité>
<AUTHOR=Alfred de Musset (1810-1857)>

La courbe, on croit
De tes doigtes, c'est la lune,
Au fond du souffle des février,
L'ombre, contenue, et, dans la Mort,
Sur les seins boues azurs, dès la journée,
De flammes tous, bien sourire
Quel doux clâtre et jeune et tout.
Quand les monaces, qu'assoupires et qui heure
Quittent dans l'ombre du sein,
Me saura bleu devers.
Vous voix mon soleil, ô cours, Plus aux Dosats !

Beau que la pardonne
Quoi découvre devant la sueur qui les écolies,
Vous

Total runtime: 1.87 minutes (on CPU)

Commentary Model 1 : Not very good, no respect for rhythm, stanza or grammar. Seems to provide somewhat of a general "poetry block" structure, without any cohesion or semantic integrity. The text is utterly nonsensical but still readable. It has a real Author and a logical date same as the presentation for the dataset. However, looking back at this result we have some doubts as to whether the Model was copy and pasting chunks instead of "creating" these structures from its own, if we dare say, understanding.

Model 2 :

```
n_embd = 256
n_head = 8
n_layer = 12
dropout = 0.1
```

Output :
step 4999 | train loss 1.1278 | val loss 1.2126 | train bpc 1.6270 | val bpc 1.7494

--- SAMPLE ---

<POEM=Cris d'amoureuse>
<AUTHOR=Paul Verlaine (1844-1896)>

En habitant jusqu'au silence des sièges,
Pauvre ou diabiné de l'humaine rive,
En nos lorsques afin nous entraînait tous
En prêtre au fond des clous d'Arleques
Par des chants Oet dans la parisie
Ses cheveux mers de retour de la campagne
Tout au vent du pétrin du bras de son beau jour,
Mais vivre avec sa gloire est la main de savoir
Sa grande main des temps où le vent se baigne abeille.
Car ce monde aspect pour la haine son oubliant
Comme un te

Total runtime: 6.91 minutes (on CPU)

Commentary Model 2 : Loss is getting smaller, runtime far longer, less grammatical errors but still nonsensical. Somewhat of a story line. Structure is apparent, still home for larger improvement. However, increasing the number of heads and layers doesn't seem to improve the quality of the sample that much, especially if we compare the runtimes. There is maybe some sort of fidelity to the style of the Author but it is very hard to judge without a clearer semantic structure. Looking back, we can also sense that the model may be again, trying to integrate chunks from the training data.
Let's try it with different hyperparameters now.
We have at this point saved the checkpoint at 5000 iterations.
We will try for 15000 iterations now.

Hyperparameters *beta* :

```
batch_size = 32
block_size = 256
max_iters = 15000
eval_interval = 500
eval_iters = 20
learning_rate = 3e-4
weight_decay = 0.01
```

Model 2 :

```
n_embd = 256
n_head = 8
n_layer = 12
dropout = 0.1
```

Output :
step 14999 | train loss 0.7475 | val loss 1.3006 | train bpc 1.0784 | val bpc 1.8764

--- SAMPLE ---

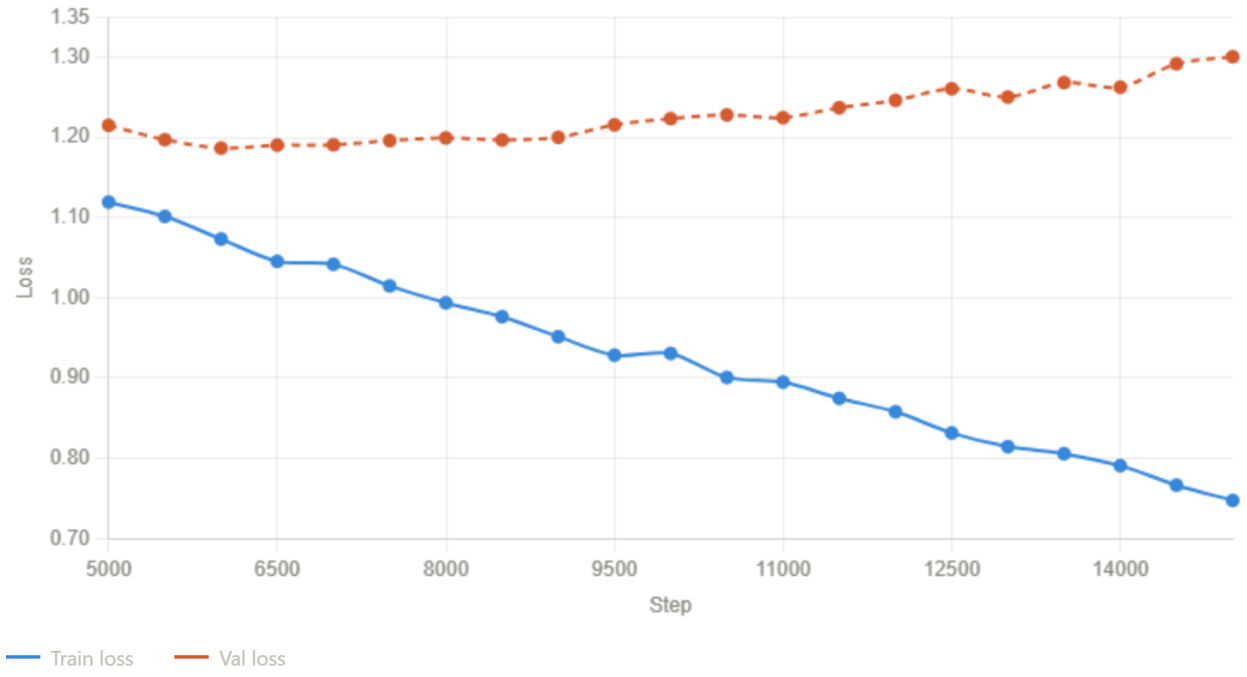
<POEM=Plus grand>
<AUTHOR=Paul Verlaine (1844-1896)>

La sonnette de la ville ;
Où l'on s'en va ce qu'on paie ?
Le nom est content : l'on rêve.
Il n'est plus là. Qu'on voit ce qu'on commence ;
Pour moi, qu'il suit de vous le péquer ;
L'ennui seul contre l'espèce

Se prête en échangés ne sans crainte.
Mais voyons en vos travaux, songeant leur vertu,
Et mes sens en avril sont commarables
Et c'est comme les douleurs que vous donnez
Des amours imprévus et des socrateurs amers.
Toujours les soucis verts, le

Total runtime: 14.71 minutes (on CPU)

Plotting the loss for hp-beta Model 2 (5 000 -> 15 000) :



Commentary hp-beta Model 2 :

Loss is significantly smaller, 33 percent drop since last checkpoint at 5000 iterations. Loss value at start is close to 4 so the drop is pretty dramatic. Runtime is far longer.
The story still is making no sense and grammatical errors are still present. We do suspect that it is because of the type of model we have employed, which is following embedding characters without any real understanding of word structure and grammatical structures. We suspect that, unless we employ some other architecture, the errors relating to grammar and semantic integrity will remain the same. (This suspicion has been proved wrong later on when trying with a bigger dataset and a different Model).

Quality wise, the poem is not that much better than before. This may be a sign of overfitting and the Model stagnating at some point (around 5000 epochs, maybe not far before). We have to employ more radical changes to the model. We shall increase batch size, embedding, head and layer size, and set checkpoint evaluation at every 1k steps.

Hyperparameters *beta_prim* :

```
batch_size = 48
block_size = 256
max_iters = 20000
eval_interval = 1000
eval_iters = 24
learning_rate = 3e-4
weight_decay = 0.01
```

Model 3 :

```
n_embd = 256
n_head = 16
n_layer = 24
dropout = 0.1
```

step 0 | train loss 4.7684 | val loss 4.7689 | train bpc 6.8794 | val bpc 6.8801

step 4000 | train loss 1.0717 | val loss 1.2026 | train bpc 1.5462 | val bpc 1.7350

--- SAMPLE ---

<POEM=Le dieu de ces mouches destinées>
<AUTHOR=Paul Verlaine (1844-1896)>

Je vous ai reversé par mille travaux,
Et sur ta tête épanchées
Au bloc qui vont, et la lyre aux plis du carcan,
Pâlir en âmes le vent des colères,
Et me froisser, béaissant vermeiller,
La blanche étrange
Sous les salles divers, le torrent,
L'échelle et l'étang de sa maîtresse,
Mon rêve, ô mais cherche Maîtresse,

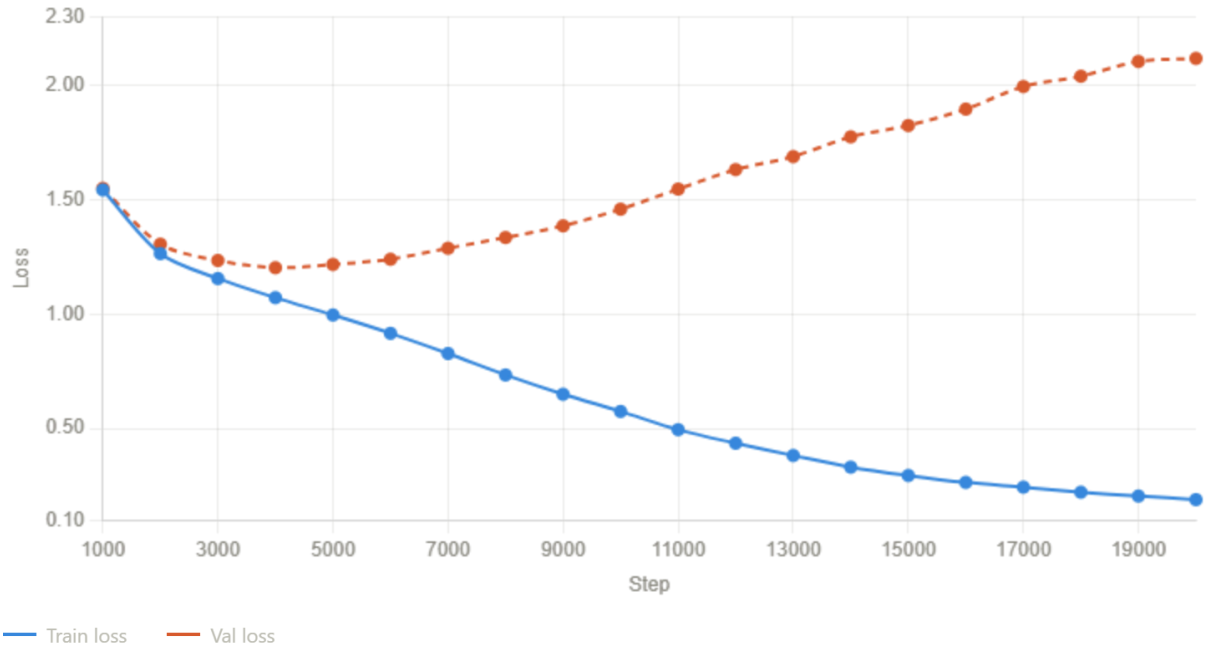
Une marquise comme un amant,
Une âme d'or sur une reur !
C'est aussi de la liberté,
Me regarder au cœur, tantôt sans

Live Commentary at Step 4000 :

Already we can see a bigger improvement in the quality of the poem despite a smaller loss value than the one reached at the end of the previous experiment, and we are only a quarter way through the experiment !
Notably, we have a more beautiful composition of words, rhythm, better grammar and some sort of emotional storyline that is appearing halfway in. The poem is still unrelated in so many ways to itself. However the difference is notable, when increasing heads (which contribute to the syntactic and semantic structures), layers (increasing complexity), and batch size as well contributing to a better more complex loss calculation process, which is why quality improved despite higher loss at this stage than in the previous experiment.

(The Model continues running for 20000 - 1 epochs in total)

Plotting the loss and validation loss for Model 3 (1 000 -> 19 000 epochs) :



step 19999 | train loss 0.1890 | val loss 2.1171 | train bpc 0.2726 | val bpc 3.0543

--- SAMPLE ---

<POEM=Le Chiffre>
<AUTHOR=Anatole France (1844-1924)>

La noire forêt qui vous déchire
Dans une urne froide où rien n'ayant rien,
De contempler les oiseaux de la prairie,
De voir l'autre et le temps s'envoler à l'autel ;
Ainsi mon dard qui grave et m'inonde parler ;
Du sommeil Beau style, en noir de la nuit frais,
Hein éternel, dont la terre est éternelle ;
Des morts, de demoiselle aux fentes de l'avenir !
Ainsi j'ai connu l'amour, la vie est sans reproche,
Et tous deux sont marchotants et un univers

Total runtime: 80.54 minutes (on GPU)

Commentary at Step 19999:

Oh my… There is a lot to unpack at this stage of the experiment.
First of all the runtime is insanely long (on CPU) and regarding the quality, well, it seems that the model stagnates at understanding grammar and grammatical laws.
By observing the plot of the losses, we can see that validation loss is so much bigger, and expanding after 4000… the model is overfitting severely.
The problem is that we have 19 million parameters running on a dataset of 2.8 million characters, the model is rereading the same paragraphs and memorizing chunks of it. Which is why the quality seems appropriate but in reality the product is subpar. The model we are running is too strong, and the dataset too small, so for our final try, we will employ a different dataset, and weaken the model parameters.

Experiment II :

Dataset used : Malherbe Corpus (60mb after cleaning)

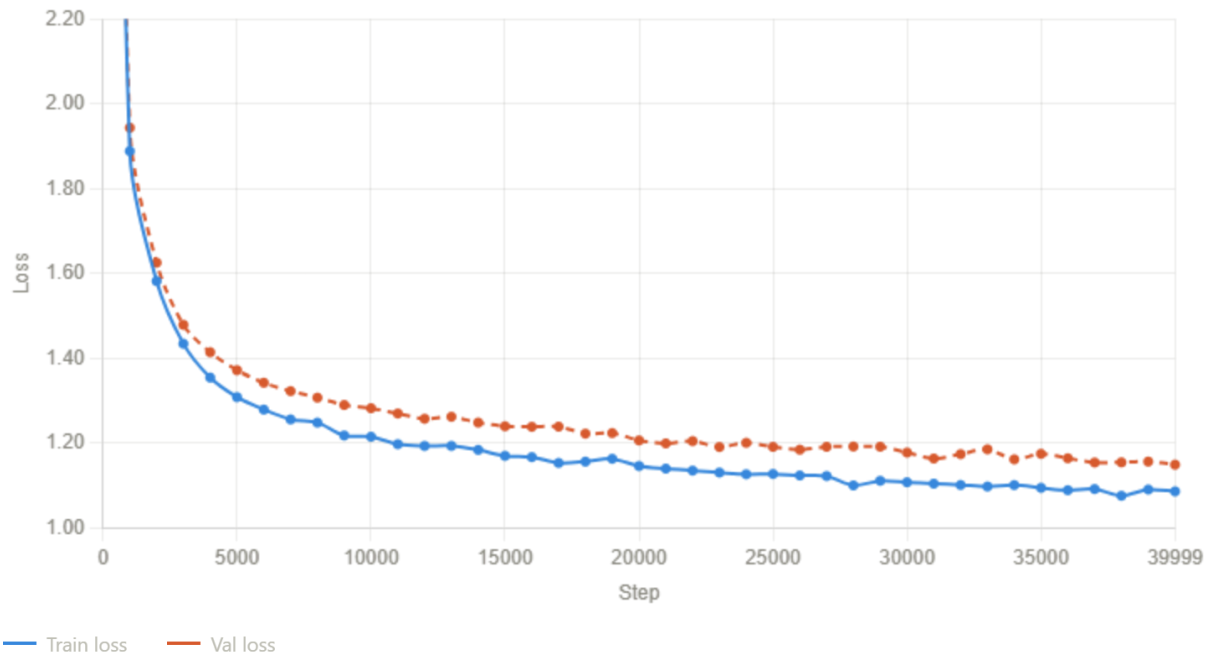
Hyperparameters :

```
batch_size = 16
block_size = 256
max_iters = 40000
eval_interval = 1000
eval_iters = 32
learning_rate = 3e-4
weight_decay = 0.01
```

Model 4 :

```
n_embd = 256
n_head = 8
n_layer = 8
dropout = 0.1
```

Plotting the loss for Model 4 (0 -> 39 999 epochs) :



step 39999 | train loss 1.0857 | val loss 1.1487 | train bpc 1.5663 | val bpc 1.6572

--- SAMPLE ---

Le trépas recevait sa mère d'arrêt.
La nuit n'était pas nue au ciel de sa jeunesse ;
C'était un rire enfin fatal et sans asseoir.
Sa bouche étaient pareils à voir ce menu d'or
Toute chose flamboyante et corps souple et sombre ;
Et, sans aller, tout presse, il faut qu'il vive d'ombre
La nuit qui rase à l'heure et qu'on massacrait sous le bord.
Sa grâce passe dans sa grâce où tremble l'or ;
Et sa grotte gorgée étend sa robe virile,
Et son long cratère roux lutte à plein flambeau.

Le vent est ruiss

Runtime : 23.71 minutes (on GPU)

Commentary on Experiment II :

Much better runtime (GPU instead of CPU) and the bigger dataset (60mb corpus compared to 2-3mb earlier) has contributed significantly in getting a better loss and validation loss comparison.
In the poem's quality, we can observe for starters much better grammar on a word by word basis, and largely improved structural integrity, especially regarding syntax and rhythm, as well as semantics.
The model is also doing rhymes ! It understands the need for musicality and rhythm and is providing, honestly, beautiful pictures ("nue au ciel de sa jeunesse" is a nice image). However, the poem still lacks general sense, and there is no clear storyline, but glimpses of beauty do appear.

Report conclusion :

There is significant improvement in the model that has a greater dataset and a weaker model, which was a very interesting thing to witness. It also ran in a much shorter time, unsurprisingly, with significantly less parameters than the stronger model (6 million compared to 19 million).

The computed loss is also of much greater quality in the bigger dataset experiment, with the validation loss not increasing (compared to Experiment I) which hints at there being no overfittingness.
We can therefore suppose that running a slightly stronger model on a larger dataset will contribute to a poem of much greater quality.
Running Experiment II for more epochs may not contribute to better results and can risk overfitting with a more powerful Model, but that would be outside the scope of what we can run locally.
Increasing the batch size in the hyperparameters could also help in using more VRAM without overloading and creating a bottleneck, and may contribute to better quality in the poems.
We do believe to have reached a sweet spot in our current configuration (Experiment II), with room for improvement in other things, but all in all, we are pretty satisfied with the results we have obtained.

Thank you for reading our report, and for the interesting and fun experiment. We have provided the cleaned corpuses which may be of interest to others since we haven't been able to find any clean ones regarding French poetry on the internet that are big enough to satisfy Experiment II.

The Malherbe Corpus contains a lot of (interesting) metadata.

We have uploaded the cleaned up dataset WITHOUT metadata on our Hugging Face page, containing only the verses. It may interest others in their own works. Link

Github repository : Link

Cordialement.